

Question Bank (Microcontroller – BEC405A)

Module 4

1. Define Interrupt. List the steps involved in executing interrupt.
2. Explain how interrupt priority can be changed using IP register. Also explain the default priorities assigned to interrupts in 8051 microcontroller.
3. Explain the bit structure of Interrupt Enable(IE) Register. Show the instruction to enable the serial interrupt, timer 0 interrupt, and external hardware interrupt 1.
4. Write a ALP that continuously get 8-bit data from P0 and sends it to P1 while simultaneously creating a square wave of 200 μ s period on pin P2.1. Use timer 0 to create the square wave. Assume that XTAL = 11.0592 MHz.
5. Explain how programming of external hardware interrupts is done in 8051 microcontrollers with a code.
6. Write a C program using interrupts to generate a square wave on port pin P1.2 of 1kHz using timer-0 in mode 2.
7. Explain programming of serial communication interrupts.

Module 5

1. Write a program to display "HELLO WORLD" by interfacing LCD display to 8051 microcontroller.
2. With a neat diagram, explain the DAC interface to 8051 and write a program to generate a triangular waveform.
3. With neat diagram explain the interfacing of ADC0804 to 8051 microcontroller and write an assembly language program to convert an analog input into its equivalent digital value and display it on port P0.
4. With neat diagram explain the interfacing of stepper motor to 8051 Microcontroller and write a C program to rotate stepper motor for one clockwise and one anticlockwise rotation.
5. With neat diagram explain the interfacing of DC motor to 8051 Microcontroller and write a C program to rotate DC motor based on the status of the switch (SW) connected to port P2.1.
If SW=0; Rotate the motor in Anticlockwise direction.
If SW=1; Rotate the motor in Clockwise direction